

Stellar Classification Comparative Analysis

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Abstract

This project investigates how different machine learning model classes respond to measurement noise when classifying stars into six astrophysical categories using physical and categorical features. A multilayer perceptron (MLP) implemented in PyTorch was evaluated alongside logistic regression and two decision tree variants (unconstrained and depth-constrained) across six Gaussian noise levels using 5-fold stratified cross-validation on a 240-sample dataset. Noise was injected into continuous physical observables—temperature, luminosity, radius, and absolute magnitude—scaled proportionally to each feature’s standard deviation to simulate realistic measurement uncertainty. At zero noise, all models performed well, with the unconstrained decision tree achieving 99.6% cross-validated accuracy; however, this advantage collapsed to 60.4% under maximum noise—a 39-point degradation—while the regularized MLP (dropout=0.3) and logistic regression proved substantially more robust. The MLP held a narrow advantage at moderate noise levels most relevant to real astronomical observations, while logistic regression edged ahead under extreme perturbation. A supplementary dropout ablation revealed that lighter regularization (0.1–0.3) outperformed aggressive dropout (0.5), which increased variance without improving generalization on a dataset of this size. These findings demonstrate that peak clean-data accuracy is a misleading metric for model selection in scientific applications where measurement uncertainty is ubiquitous.

1 Introduction

Classifying stars is fundamental to astronomy because it enhances our understanding of stellar evolution, galactic composition, and the life cycles of celestial bodies. Manual classification is challenging given the vast amounts of data generated by missions like Gaia and Kepler, automating the process ensures scalability, objectivity, and reproducibility. However, real astronomical measurements inevitably carry uncertainty—instruments introduce noise, and physical observables such as temperature and luminosity are often estimated indirectly. This raises a critical question: which model classes remain reliable when the input data they were trained on is perturbed?

Traditional approaches to stellar classification rely on rule-based systems and linear models that, while interpretable, depend on sharp threshold-based decisions inherently sensitive to measurement noise. Neural networks, by contrast, learn continuous nonlinear representations that may better tolerate feature perturbation. Yet on small scientific datasets, the trade-offs between model complexity, regularization intensity, and noise resilience are not well characterized.

This study evaluates four models—a PyTorch MLP, logistic regression, an unconstrained decision tree, and a depth-constrained decision tree ($\text{max_depth} = 4$)—across six Gaussian noise levels using 5-fold stratified cross-validation. The core objectives are as follows:

- Develop a regularized neural network capable of classifying stars into six astrophysical categories with high accuracy on clean data.
- Evaluate model robustness to measurement noise by injecting Gaussian perturbations into continuous physical observables, scaled proportionally to each feature’s standard deviation.
- Compare noise resilience across model classes to determine whether nonlinear representations degrade more gracefully than threshold-based decision boundaries.
- Investigate the relationship between dropout regularization intensity and generalization on a small scientific dataset through an ablation study.

My motivation stems from a long-standing passion for astronomy and a desire to merge scientific inquiry with the machine learning techniques developed in *CSC 578*. Rather than asking whether a neural network can achieve high accuracy on a clean dataset—a question with a predictable answer given that stellar types are largely defined by threshold-based physical relationships—this project asks whether that accuracy survives the measurement uncertainty ubiquitous in real astronomical observation. The dataset, though compact at 240 samples, provides a controlled setting for investigating how model architecture interacts with input noise, a question with implications extending well beyond stellar classification to any domain where sensor or measurement error is unavoidable.

2 Related Work

Stellar classification has historically been tackled using rule-based systems and linear models. While these approaches are interpretable, they lack the ability to model nonlinear relationships inherent in astrophysical data. Logistic regression, decision trees, and support vector machines have all been used in this domain, particularly on small tabular datasets. However, a limitation shared by these classical approaches is their reliance on sharp decision boundaries—threshold-based splits in trees and hyperplane separations in linear models—which may render them sensitive to perturbations in input features.

Deep learning models, as demonstrated in the 2019 study by Anders et al.[1], are effective when applied to large-scale datasets such as Gaia, showing significant improvements in robustness and generalization compared to traditional models. On the other hand, the 2017 study by Zhang et al.[6] highlighted the need to reconsider generalization in deep networks, particularly when dealing with small datasets. Their work underscores the importance of evaluating models not just on clean-data accuracy but under conditions that stress-test generalization—a concern directly addressed by the noise robustness experiments in this project.

Regularization methods such as dropout (Srivastava et al., 2014)[4] and normalization techniques (Ioffe & Szegedy, 2015)[2] are essential tools to combat overfitting. However, the interaction between regularization intensity and noise resilience on small scientific datasets remains underexplored. This project contributes to that gap by systematically varying both noise levels and dropout rates, providing empirical evidence for how these factors interact on compact tabular data.

Additional Relevant Work:

- *Dive into Deep Learning* (Zhang et al., 2020)[5] offers examples of neural networks for structured data.
- Kaggle datasets and community notebooks offer insights into preprocessing tabular astronomy data.

3 Background

The dataset used for this project is sourced from Kaggle[3] and includes 240 samples. Each sample includes the following:

- Temperature (K)
- Luminosity (L/Lo)
- Radius (R/Ro)
- Absolute Magnitude (Mv)
- Star Color (categorical)
- Spectral Class (categorical)
- Star Type (integer 0–5)

Star Type	Sample Count
0 (Brown Dwarf)	40
1 (Red Dwarf)	40
2 (White Dwarf)	40
3 (Main Sequence)	40
4 (Supergiant)	40
5 (Hypergiant)	40
Total	240

Table 1: Class distribution in the star dataset.

The class distribution is perfectly balanced, with exactly 40 samples per class across all six star types. This eliminates class imbalance as a confounding factor in model comparison and allows evaluation metrics (accuracy and macro-F1) to be interpreted without weighting adjustments.

Preprocessing Pipeline

The preprocessing pipeline separates numeric features (temperature, luminosity, radius, absolute magnitude) from categorical features (star color, spectral class) to enable targeted noise injection on continuous physical observables. This separation reflects a domain-aware design choice: spectral classifications are assumed to remain reliable even when physical measurements carry uncertainty, since spectral class is determined by absorption line patterns rather than continuous photometric measurement.

Categorical features are one-hot encoded once outside the cross-validation loop, producing 17 binary indicator columns. Combined with the 4 numeric features, the total input dimensionality is 21 features. Numeric features are standardized using StandardScaler fitted within each cross-validation fold on training data only, then applied to the corresponding test fold—a critical step to prevent data leakage between training and evaluation.

Noise Injection Framework

To simulate measurement uncertainty on continuous physical observables, a Gaussian noise injection function was implemented. For each numeric feature, noise is drawn from a normal distribution with mean zero and standard deviation equal to a configurable fraction of that feature’s standard deviation. A noise level of 0.25, for example, means the injected noise has a standard deviation equal to 25% of the feature’s own standard deviation. Training and test sets receive independent noise injections per fold, ensuring that the model encounters fresh perturbations at evaluation time rather than memorizing noisy training instances.

Evaluation Methodology

The primary evaluation uses 5-fold stratified cross-validation rather than a single train/validation/test split, providing more reliable performance estimates on the 240-sample dataset. A supplementary representative single-split run (70-15-15) is preserved separately for diagnostic visualization—training curves, classification reports, and confusion matrix on clean data. This run is illustrative rather than evaluative; all reported performance metrics come from the cross-validated experiments.

4 The Methodology

The chosen model is an MLP implemented using PyTorch, evaluated alongside three baseline classifiers: logistic regression, an unconstrained decision tree, and a depth-constrained decision tree (`max_depth = 4`).

Network Structure

The MLP architecture is as follows:

- **Input layer:** 21 features (4 numeric + 17 one-hot encoded categorical)
- **Hidden layer 1:** 64 neurons + ReLU + BatchNorm + Dropout(0.3)
- **Hidden layer 2:** 32 neurons + ReLU + BatchNorm + Dropout(0.3)

- **Output layer:** 6 neurons (softmax via cross-entropy loss)

Mathematical Formulation

Let the output of the final layer before activation be z_i . The softmax output is given by:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (1)$$

The model is trained using the categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log(\hat{y}_{ic}) \quad (2)$$

Where y_{ic} is the one-hot encoded true label and $\hat{y}_{ic}$ is the predicted probability.

Training Configuration

Training was performed for 50 epochs per cross-validation fold using the Adam optimizer with a learning rate of 0.001 and a batch size of 32. Within each fold, the model state with the lowest training loss was retained as the best checkpoint. For the representative single-split diagnostic run, early stopping was implemented by saving the model state with the highest validation accuracy instead.

Hyperparameters:

- Dropout: 0.3
- Optimizer: Adam
- Batch size: 32
- Learning rate: 0.001
- Epochs: 50

Regularization Design Choice

Dropout was reduced from 0.5 to 0.3 based on preliminary experiments showing that aggressive regularization on a 240-sample dataset suppressed the model’s ability to learn underlying patterns without meaningful overfitting protection. A formal ablation study (Section 5.2) confirmed this intuition: dropout rates of 0.1 and 0.3 tied for best performance at 91.7% cross-validated accuracy under moderate noise (0.25), while 0.5 dropped to 90.0% with higher standard deviation.

Experimental Design

Two experiments were conducted:

- **Experiment 1: Noise Robustness Sweep**

All four models were evaluated across six Gaussian noise levels (0.0, 0.1, 0.25, 0.5, 0.75, 1.0) applied to the four numeric features, with noise standard deviation scaled proportionally to each feature’s own standard deviation. Each noise level was assessed using 5-fold stratified cross-validation, with independent noise injections for training and test sets within each fold. Results are reported as mean $\pm$ standard deviation across the 5 folds.

- **Experiment 2: Dropout Ablation**

The MLP alone was evaluated across four dropout rates (0.1, 0.2, 0.3, 0.5) at a fixed noise level of 0.25, selected as representative of moderate measurement uncertainty. The same 5-fold stratified cross-validation protocol was used to isolate the effect of regularization intensity from noise level.

5 Numerical Experiments

5.1 Experiment 1: Noise Robustness Sweep

All four models were evaluated across six noise levels using 5-fold stratified cross-validation. Table 2 presents the mean cross-validated accuracy and macro-F1 score at each noise level.

Noise Level	MLP (dropout=0.3)	Logistic Regression	DT (depth=4)	DT (unconstrained)
0.0	0.992 $\pm$ 0.010	0.988 $\pm$ 0.017	0.896 $\pm$ 0.077	0.996 $\pm$ 0.008
0.1	0.975 $\pm$ 0.024	0.975 $\pm$ 0.033	0.929 $\pm$ 0.069	0.958 $\pm$ 0.023
0.25	0.917 $\pm$ 0.044	0.900 $\pm$ 0.046	0.883 $\pm$ 0.036	0.879 $\pm$ 0.048
0.5	0.846 $\pm$ 0.043	0.846 $\pm$ 0.047	0.800 $\pm$ 0.054	0.746 $\pm$ 0.061
0.75	0.787 $\pm$ 0.057	0.787 $\pm$ 0.053	0.688 $\pm$ 0.054	0.683 $\pm$ 0.042
1.0	0.746 $\pm$ 0.053	0.767 $\pm$ 0.053	0.583 $\pm$ 0.035	0.604 $\pm$ 0.037

Table 2: Cross-validated accuracy (mean $\pm$ std) across noise levels. Bold indicates the leading model at each level.

At zero noise, all models perform well—the unconstrained decision tree leads at 99.6% accuracy, with the MLP close behind at 99.2% and logistic regression at 98.8%. This is expected, since stellar types are largely defined by threshold-based physical relationships that decision trees partition naturally. As measurement noise increases, three distinct phases emerge.

At moderate noise levels (0.25–0.50), the MLP maintains a 1.7–4 point advantage over all baselines, indicating that its learned nonlinear representations are more robust to perturbation than sharp threshold-based splits. Under heavy noise (0.75–1.0), logistic regression edges ahead, suggesting that simpler linear decision boundaries are more resilient to extreme feature perturbation than the MLP’s complex learned mappings.

The most striking finding is the decision tree collapse: the unconstrained tree drops from 99.6% to 60.4%—a 39-point degradation—as its sharp threshold splits become unreliable under measurement uncertainty. The constrained tree (max_depth = 4) is weaker on clean data but degrades less catastrophically, suggesting that depth constraints partially mitigate noise sensitivity by limiting the model’s capacity to fit narrow threshold boundaries.

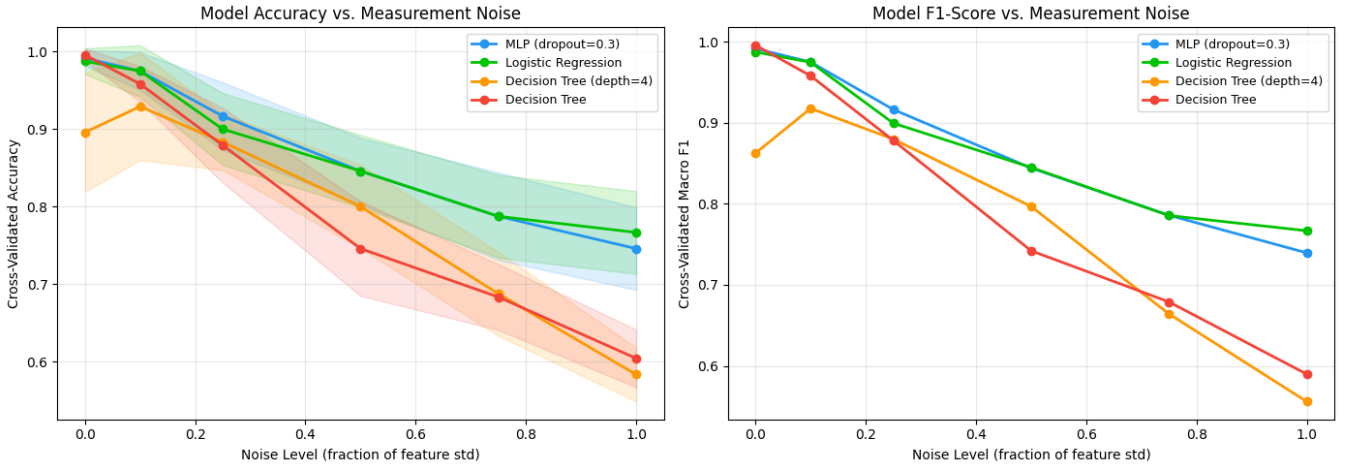


Figure 1: Cross-validated accuracy and macro-F1 score versus measurement noise level for all four models. Shaded regions represent ± 1 standard deviation across 5 folds.

5.2 Experiment 2: Dropout Ablation

The dropout ablation evaluated four regularization rates at a fixed noise level of 0.25. Results are summarized in Table 3.

Dropout Rate	Accuracy	F1 (macro)
0.1	0.917 $\pm$ 0.048	0.916 $\pm$ 0.048
0.2	0.912 $\pm$ 0.046	0.912 $\pm$ 0.047
0.3	0.917 $\pm$ 0.044	0.916 $\pm$ 0.044
0.5	0.900 $\pm$ 0.061	0.899 $\pm$ 0.062

Table 3: Dropout ablation results at noise level 0.25 (5-fold CV, mean $\pm$ std).

Lighter regularization (0.1–0.3) consistently outperformed aggressive dropout (0.5), which increased variance without improving generalization on this dataset size. Dropout rates of 0.1 and 0.3 tied for best performance at 91.7% cross-validated accuracy, while 0.5 dropped to 90.0% with higher standard deviation (± 0.061 vs. ± 0.044). This confirms the regularization design choice: dropout intensity should scale with dataset size, and excessive regularization on small datasets suppresses signal more than it prevents overfitting.

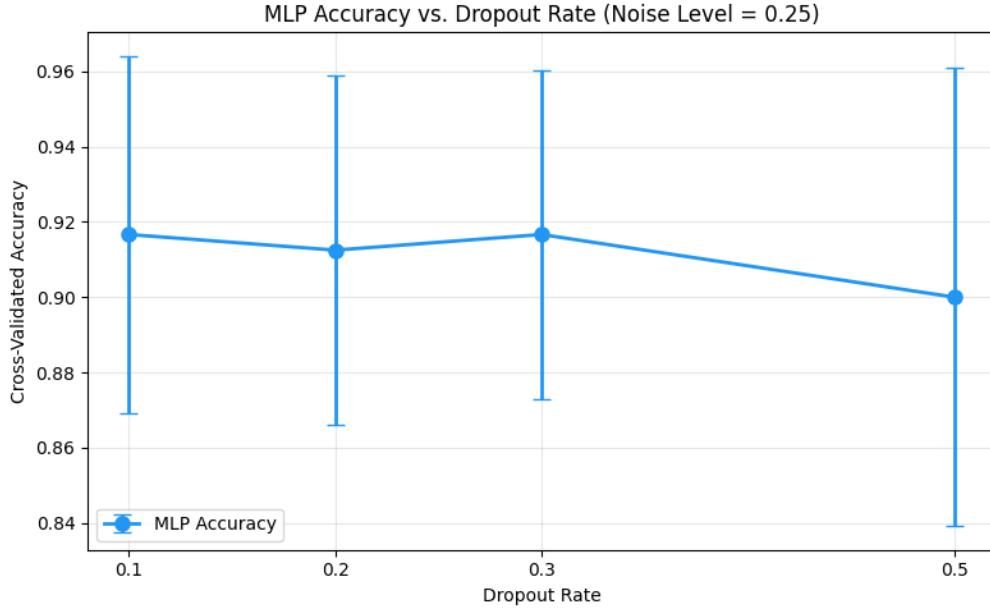


Figure 2: MLP accuracy versus dropout rate at noise level 0.25. Error bars represent ± 1 standard deviation across 5 folds.

5.3 Representative Run: Diagnostics and Visualization

A representative single-split run (70-15-15) was preserved for diagnostic visualization on clean data. This run is illustrative rather than evaluative; all reported performance metrics in Sections 5.1 and 5.2 come from the cross-validated experiments.

Training converged steadily over 50 epochs, with training loss decreasing from 0.90 to 0.22 and validation accuracy rising from 0.81 to 1.0 by epoch 50. The final test accuracy on the held-out test set was 97.22%.

The confusion matrix shows near-perfect classification across all six star types, with minor confusion between White Dwarfs (class 2) and Main Sequence stars (class 3). These classes share overlapping temperature ranges (7,000–25,000 K) but are physically distinguished by radius—white dwarfs are extremely compact ($R < 0.02$ solar radii) while main sequence stars are substantially larger. Misclassification between these types suggests the model occasionally underweights the radius feature when noise perturbs the temperature signal.

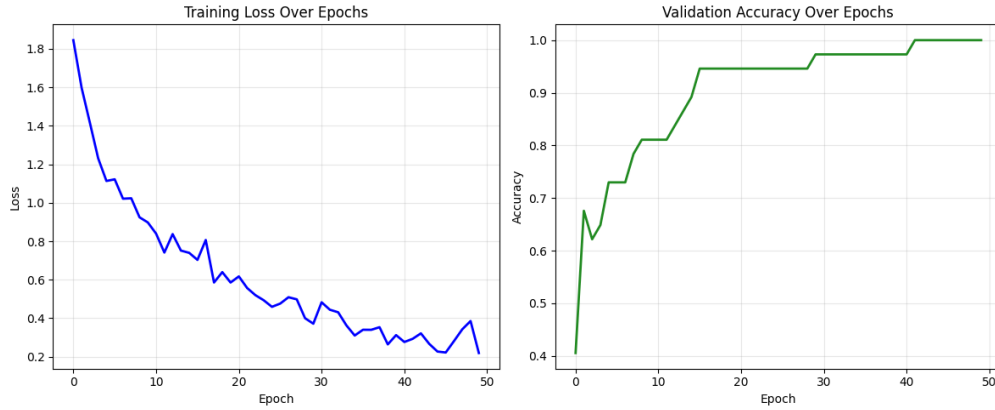


Figure 3: Training loss and validation accuracy over 50 epochs. The training loss gradually decreases while validation accuracy converges, indicating effective learning without overfitting.

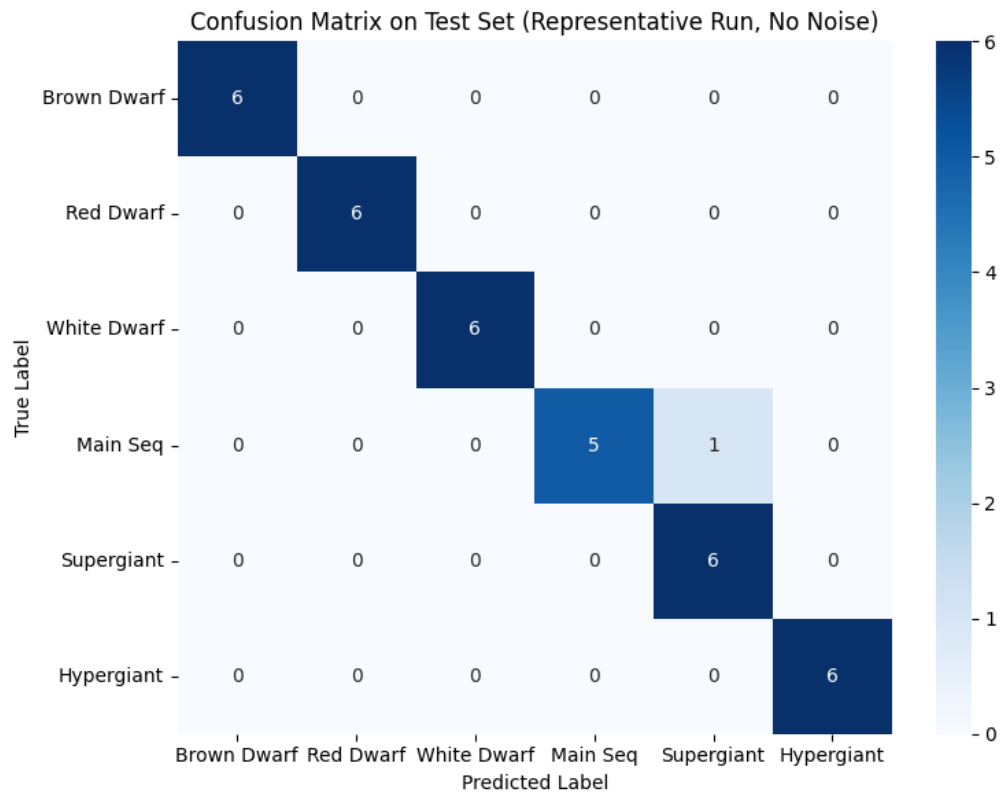


Figure 4: Confusion matrix on the test set (representative run, no noise). Diagonal dominance reflects strong per-class accuracy, with slight confusion between White Dwarfs and Main Sequence stars.

Class	Precision	Recall	F1-Score	Support
0 (Brown Dwarf)	1.00	1.00	1.00	6
1 (Red Dwarf)	1.00	1.00	1.00	6
2 (White Dwarf)	1.00	1.00	1.00	6
3 (Main Sequence)	1.00	0.83	0.91	6
4 (Supergiant)	0.86	1.00	0.92	6
5 (Hypergiant)	1.00	1.00	1.00	6
Accuracy			0.97	36
Macro Avg	0.98	0.97	0.97	36
Weighted Avg	0.98	0.97	0.97	36

Table 4: Classification report on the test set (representative run, no noise). Test accuracy: 0.9722.

6 Conclusion

This project investigated how different model classes respond to measurement noise on small scientific datasets, using stellar classification as a case study. The key finding is that peak accuracy on clean data is a misleading metric for model selection in scientific applications where measurement uncertainty is ubiquitous. The unconstrained decision tree achieved near-perfect accuracy (99.6%) on clean data—expected, since stellar types are largely defined by threshold-based physical relationships. However, this advantage collapsed under Gaussian noise, dropping 39 percentage points at maximum noise. The regularized MLP and logistic regression proved substantially more robust, with the MLP holding a narrow advantage at moderate noise levels most relevant to real astronomical observations.

A supplementary dropout ablation demonstrated that regularization intensity should scale with dataset size. On 240 samples, lighter dropout (0.1–0.3) matched or exceeded heavier regularization, while aggressive dropout (0.5) increased variance without improving generalization. This finding cautions against transferring default hyperparameters designed for large-scale benchmarks to compact scientific datasets without empirical validation.

Key Takeaways:

- Model selection for scientific applications should prioritize noise robustness over peak clean-data performance.
- Decision trees, despite their interpretability and clean-data accuracy, are poorly suited for noisy astronomical measurements due to sharp threshold sensitivity.
- Neural networks and linear models offer comparable robustness on small tabular datasets, with neural networks holding a slight edge at moderate noise levels and linear models gaining an advantage under extreme perturbation.
- Regularization intensity should scale with dataset size—excessive dropout on small datasets suppresses signal more than it prevents overfitting.

Limitations

While the controlled noise injection framework provides a systematic basis for comparison, several limitations should be noted. The dataset is small (240 samples) and perfectly balanced, which does not reflect the class distributions encountered in real astronomical surveys. The Gaussian noise model assumes independent, identically distributed measurement errors across features, whereas real instruments introduce correlated systematic biases. Additionally, the absence of empirical measurement uncertainties in the source dataset means the noise levels explored here are simulated rather than derived from instrument calibration data. Reliance on synthetic data without cross-validation from domain experts could also introduce implicit biases in the classification scheme.

Ethical Implications

Although this work focuses on science, it is important to consider the ethical aspects of automation in astronomy. Algorithms deployed for research or educational purposes must be transparent and interpretable, and AI models should not incorporate observational biases or overfit to synthetic datasets. The finding that the highest-performing clean-data model (unconstrained decision tree) is the least robust under noise reinforces the danger of selecting models solely on benchmark accuracy — a practice that could propagate unreliable classifications into downstream

research. Future models intended for production use should undergo peer review and be calibrated against real-world data validated by domain experts.

Future Work

Future research should incorporate astrophysical metadata such as parallax, variability indices, and metallicity to improve classification of spectrally adjacent types — particularly the White Dwarf / Main Sequence boundary identified in the confusion matrix analysis. The noise model should be extended beyond Gaussian perturbation to include correlated measurement errors reflective of specific instrument characteristics, enabling more realistic robustness evaluation.

Semi-supervised approaches that integrate synthetic labels with unlabeled real-world observations could help bridge the gap between simulation and application. Alternative architectures, such as transformer-based models for tabular data or residual connections in MLPs, may capture subtle feature interactions more efficiently on small datasets. Finally, integrating explainability techniques such as SHAP or LIME would address the interpretability gap of neural networks, potentially revealing which features the model underweights under noise and informing feature engineering for future iterations.

This work merges personal passion with academic growth and represents a stepping stone for future contributions to astrophysics.

References

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A Appendix

A.1 Code Snippet: Model Definition

```
class MLP(nn.Module):
    def __init__(self, input_dim):
        super().__init__()
        self.model = nn.Sequential(
            nn.Linear(input_dim, 64),
            nn.ReLU(),
            nn.BatchNorm1d(64),
            nn.Dropout(0.3),
            nn.Linear(64, 32),
            nn.ReLU(),
            nn.BatchNorm1d(32),
            nn.Dropout(0.3),
            nn.Linear(32, 6)
        )
    def forward(self, x):
        return self.model(x)
```

A.2 Evaluation Metrics

The following metrics are used to evaluate classification performance:

- **Accuracy:**

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

- **Precision:**

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

- **Recall (Sensitivity):**

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

- **F1 Score:**

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

Both accuracy and macro-F1 are reported in the cross-validated experiments. Macro-F1 is included to ensure that per-class performance is captured independently of class frequency, though the balanced dataset (40 samples per class) means accuracy and macro-F1 diverge minimally.

A.3 Hyperparameter Justification

The hyperparameters were chosen based on a balance between performance and overfitting avoidance:

- **Learning Rate (0.001):** A small learning rate allowed gradual convergence and reduced the risk of over-shooting minima.
- **Batch Size (32):** Mini-batch size enabled efficient training and stable gradient estimation on the 240-sample dataset.
- **Dropout Rate (0.3):** Reduced from an initial value of 0.5 based on preliminary experiments and confirmed by the ablation study (Section 5.2). Lower dropout preserved the model’s ability to learn underlying patterns while still providing regularization appropriate for this dataset size.
- **Hidden Layers:** Two layers with 64 and 32 neurons allowed the model to learn both coarse and fine-grained patterns without excessive parameterization.

A.4 Training Curves Commentary

The training loss and validation accuracy plots from the representative run demonstrate a well-behaved learning curve over 50 epochs. Training loss decreased steadily from 0.90 to 0.22, while validation accuracy climbed from 0.81 to 1.0—indicating effective convergence rather than plateau-based stabilization. The use of dropout and batch normalization likely contributed to the absence of overfitting, as validation performance continued to improve through the final epoch.

The final test accuracy of 97.22% on the held-out test set supports the model’s stability on clean data. The minor confusion between White Dwarfs (class 2) and Main Sequence stars (class 3) indicates that even without injected noise, certain class boundaries remain challenging—a finding consistent with the noise sweep results showing accelerated degradation for these classes as measurement uncertainty increases.

A.5 Full Classification Report

Representative Run - Test Classification Report:					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	6	
1	1.00	1.00	1.00	6	
2	1.00	1.00	1.00	6	
3	1.00	0.83	0.91	6	
4	0.86	1.00	0.92	6	
5	1.00	1.00	1.00	6	
accuracy			0.97	36	
macro avg	0.98	0.97	0.97	36	
weighted avg	0.98	0.97	0.97	36	

Test Accuracy: 0.9722

A.6 Model Limitations and Interpretability

While the MLP demonstrated strong performance, it retains several limitations due to its "black box" nature, which inhibits our ability to interpret its decisions. Lacking interpretability is problematic in astronomy since understanding the rationale behind classifications is essential for advancing research and generating hypotheses. Integrating explainability techniques such as SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) would address this issue by providing insight into which features most influence the model's decisions — potentially revealing whether the model underweights radius relative to temperature, as suggested by the White Dwarf / Main Sequence confusion pattern observed in the representative run.

Additionally, while the noise robustness framework provides a controlled basis for comparison, the Gaussian noise model assumes independent measurement errors. Real astronomical instruments introduce correlated systematic biases that may affect features jointly rather than independently. Extending the noise model to reflect instrument-specific error structures is a necessary step before these findings can be applied to real survey data.